

# HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

## PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and commissioned **bespoke automation control systems** using **Java Spring Boot** and **REST APIs** for enterprise manufacturing and operational processes, delivering **4+ production systems** across banking and ERP domains with **99.5% uptime SLA** through real-time monitoring with **Splunk, CloudWatch, and Dynatrace**.
- Assessed existing technologies and processes by analyzing **system architectures, data flows, and operational workflows** using **SQL (PostgreSQL, MySQL, Oracle)** and **Python scripting**; documented findings and managed change control across interconnected systems, reducing post-deployment incidents by **40%**.
- Worked within diverse teams to deliver best-in-class automation solutions by collaborating with customers on technical requirements, translating business needs into control-system specifications, and providing technical support during UAT and post-go-live phases across **8+ cross-functional automation projects**.
- Supported automation operations with technical queries and incident resolution, responding to **P1/P2 production issues** within **SLA windows** using **Linux shell scripting, log analysis, and system monitoring dashboards**; documented root causes and implemented preventive controls via **Jenkins CI/CD pipelines**.
- Engineered software for bespoke machines and manufacturing systems using **Docker containerisation, AWS (EC2, S3, Lambda), and Git branching** to enable rapid deployment and scalability; integrated automation control logic with enterprise business systems via **REST API integrations and database-level data flows**.
- Applied knowledge of robotics, safety standards, and regulatory compliance (GDPR, audit trails) in production automation environments; collaborated with electrical, mechanical, and robotics teams to ensure system reliability, safety certification readiness, and operational best practices across **12+ automated machinery deployments**.

## SKILLS

**Automation & Control Systems** – bespoke automation control systems development, machine commissioning, robotics integration, safety standards compliance, operational support.

**Software Development** – Java Spring Boot, REST APIs, Python scripting, Flask, FastAPI, Docker containerisation, CI/CD (Jenkins, Git).

**Databases & Data Systems** – SQL (PostgreSQL, MySQL, Oracle), data modelling, ETL pipelines, data warehousing, ad-hoc query development.

**Cloud & Infrastructure** – AWS (EC2, S3, Lambda, CloudWatch), Linux system administration, shell scripting, log analysis, cron job automation.

**Monitoring & Observability** – Splunk, Dynatrace, CloudWatch dashboards, SLA management, incident response (P1/P2), system reliability engineering.

**Project Management & Documentation** – Jira ticketing, Confluence documentation, change control management, Agile/Scrum, UAT coordination, cross-functional collaboration.

**Domain Expertise** – payments operations (SWIFT MT/MX, ISO 20022, SEPA), ERP systems integration, financial services regulatory compliance (GDPR), enterprise business systems.

## EDUCATION AND TRAINING

## PROJECTS

### **Autonomous Aqua Drone Navigation Control System (HCL Technologies)**

- Architected and developed a **distributed automation control system** for autonomous aqua drone navigation using **Java Spring Boot microservices** and **REST API integrations**, enabling real-time command execution and sensor telemetry processing across **50+ deployed units**.
- Implemented **Docker-based containerisation** and **AWS Lambda functions** for dynamic task scheduling and **multi-region deployment**; automated system health monitoring via **CloudWatch dashboards** and **Dynatrace APM**, achieving **99.7% uptime** and reducing operational incident resolution time by **35%**.
- Integrated robotics subsystems (GPS, sonar, propulsion) with central control logic using **Python automation scripts** and **SQL-backed telemetry databases (PostgreSQL)**; documented all system specifications, API contracts, and change control procedures in **Confluence** for cross-team alignment.
- Collaborated with mechanical, electrical, and robotics engineering teams to assess technology choices, validate safety standards compliance, and commission **5 production drone fleets**; provided post-go-live technical support and resolved **12+ critical operational issues** within SLA.
- Optimized deployment pipeline by building **Jenkins CI/CD automation** with **Git branching strategy**, reducing time-to-production by **45%** and enabling rapid iteration cycles for firmware and control-logic updates across the drone fleet.

### **Pharmaceutical Manufacturing Automation Platform & Change Control Framework (HCL Technologies)**

- Developed bespoke **manufacturing automation control systems** for pharmaceutical production line orchestration, integrating **ERP modules and enterprise business systems** with real-time machine control logic via **REST APIs and database integrations** across **Oracle, MySQL, and PostgreSQL**.
- Established a **change control management and documentation framework** for automation system updates, UAT coordination, and post-implementation support; created **Confluence runbooks and technical guides** reducing operator query response time by **50%** and ensuring regulatory compliance (GDPR, audit trails).
- Assessed existing manufacturing processes and identified automation opportunities using **Python data analysis and SQL workflow validation**; delivered **8 process optimizations** that increased throughput by **22%** and reduced manual intervention points by **60%**.

## ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)